

2023-2024
3rd year /Electronic III
Electronic and Telecommunication Department.
Engineering Science/ Kufa University

Microelectronic circuit analysis and design

Donald A. Neamen 4th edition

Electronic Design Lab course

Electronic Design into PCB production

The aim of this course is to provide students with the information and abilities they need to undertake small and large projects as well as to solve complex technological difficulties. This is achieved by possessing the knowledge necessary to create, evaluate and operate different types of electronic circuits.

Key words include:

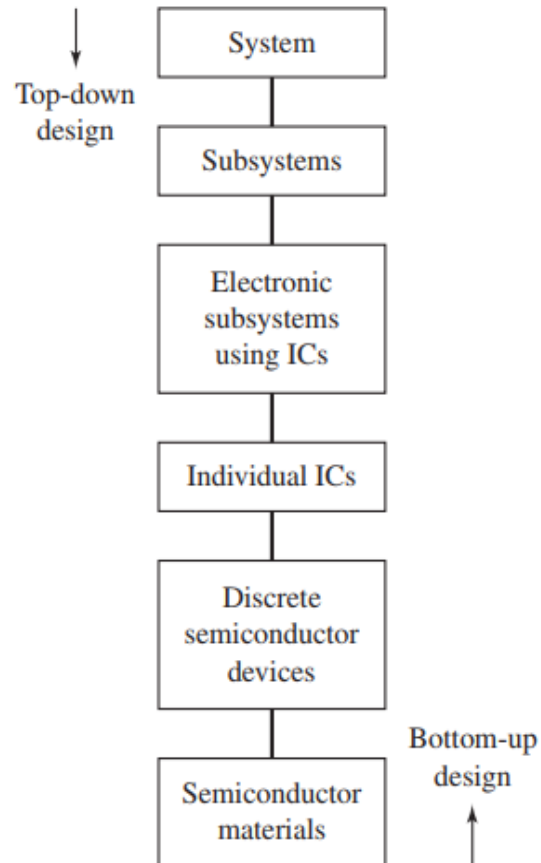
- Product Development
- PCB Design and CAD Tools
- Testing
- Management of Processes

ELECTRONIS DESIGN LAB

1. Design Electronic Circuit Lab (10).10%
 - Schematic
 - Simulations
 - PCB layout
2. Assignments/homework (2). 10%
3. Mini Project. 10%
4. Midterm Exam. 20%

Electronic Design

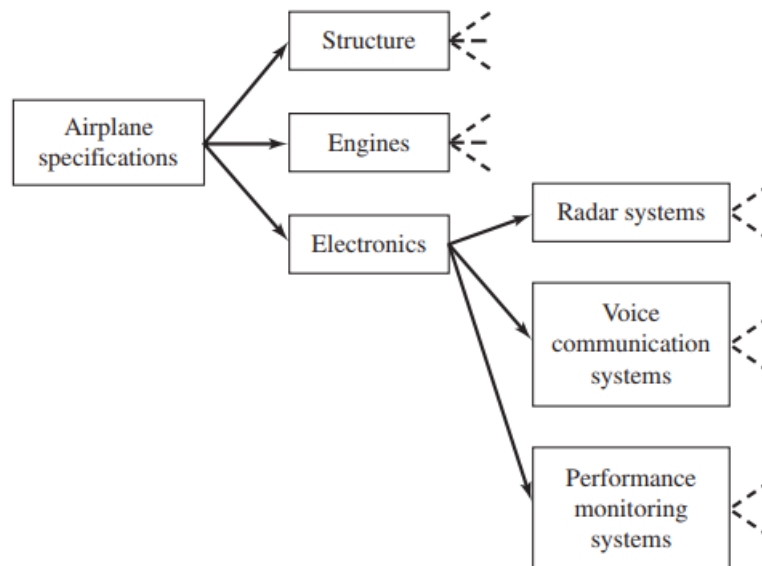
Design Approach:



1. The top-down approach is usually based on existing technologies and devices, meaning that electronic subsystems are typically designed using existing integrated circuits.
 - The design engineer must be able to evaluate existing design strategies and techniques to determine whether they are capable of achieving design or performance objectives.

- A deep knowledge of the operation and characteristics of basic circuits is essential for circuit design, and for the ability to make appropriate choices in the top-down design process.
2. The bottom-up design process typically begins in a research laboratory with the development of new and unique semiconductor materials.
- These new devices may be integrated with the entire system and lead to the development of new systems based on the properties of new devices, such as high-performance JFETs and improved optoelectronic devices.

System Design:



- The top-down approach is like a new plane. A large overall aircraft system can be divided into subsystems, as shown above.

- Electronics subsystem specifications are usually determined by the overall system specifications and as work progresses, the overall system or subsystem specifications may be improved or modified.
- During design, problems may arise, and there may be trade-offs between aircraft performance and cost. High performance may require expensive electronics.

Electronic Design

A flowchart of the general electronic design process can be seen in the figure below. Specific features of many various types of circuits, such as input and output impedances, gain characteristics, and bandwidth, must be recognized in order to specify a particular circuit configuration.

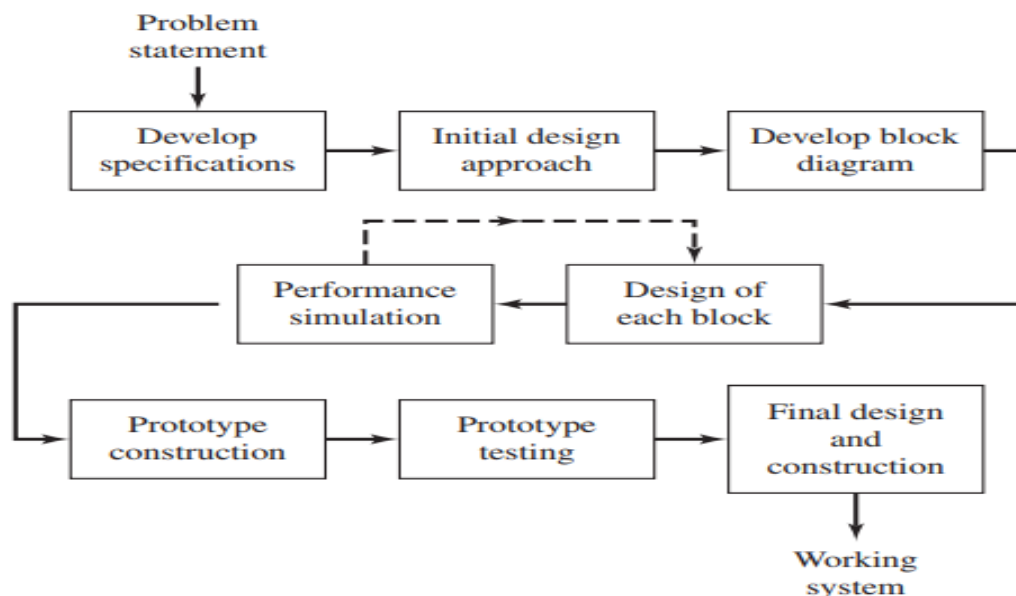


Figure PR2.3 Flowchart of the design process

The figure below shows the proposed circuit configuration for the temperature indicator design.

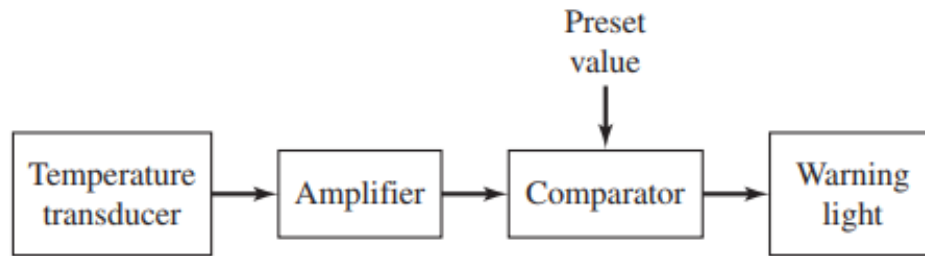


Figure PR2.4 Block diagram of a temperature warning light circuit

Amplifier design begins with the expected output signal of the temperature transducer, which in turn includes the desired level of input impedance and signal gain for the amplifier.